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Viscoelastic-adhesive modeling of hybrid adhesively bonded double-lap joints: analytical and numerical studies

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ABSTRACT

There are a limited number of studies examining the viscoelastic and viscoplastic behavior of adhesive joints. If there is more than one adhesive in the overlap region, the problem becomes more complex and viscoelastic modeling of the overlap region becomes difficult. There is no study in the literature that covers such a bonding (overlap) scheme and examines the viscoelastic behavior of the double lap joint (DLJ). In this study, the effect of viscoelasticity of polymeric adhesives on shear stress and strain distributions in the hybrid-adhesive layer of a DLJ is studied. Maxwell-Wiechert (MW) viscoelastic material model is used to define the viscoelastic behavior of the adhesives. Analytical and finite element (FE) results are compared for both mono- and hybrid-adhesive joints under the step load. Good matches are observed between the analytical and numerical results. Four different hybrid-adhesive joints with different combinations of the adherend thicknesses are investigated by considering only the analytical method. In order to investigate the effect of the loading type on the viscoelastic behavior, step and ramp loads were applied to the joint separately. It is observed that applying the ramp load to the hybrid-adhesive joints affects the shear stresses occurred at the ends of the overlap more than the mono-adhesive joints.

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1. Introduction

Nowadays, adhesives are frequently used to obtain reliable, durable, and lightweight constructions in the fields of automotive, marine, aviation, military, etc. Using adhesives by considering their advantages provides many conveniences in practice. The simplest analytical model of the single lap joint (SLJ) was first proposed by Volkersen.^[1] Raphael's pioneering model reveals a technique for varying the adhesive rigidity along the overlap length, so that a flexible adhesive carries the strain at the joint edges, while a brittle adhesive is used in the center of the overlap.^[2] A technique similar to Raphael's technique was also suggested by da Silva and Lopes.^[3] The technique is based on the approach of dividing the overlap region into three parts. In the

related study, SLJs were manufactured and tested maintaining the same brittle adhesive in the middle of the overlap, and using three different ductile adhesives of increasing ductility at the ends of the overlap. They concluded that the hybrid-adhesive technique gives joint strength improvements in relation to a brittle adhesive alone in all cases studied; and for a mixed adhesive joint to be stronger than the brittle adhesive, and the ductile adhesive used individually, the load carried by the brittle adhesive must be higher than that carried by the ductile adhesive. Tsai and Morton investigated the mechanics of double lap joint (DLJ) with unidirectional and quasi-isotropic composite adherends under tensile loading, experimentally using moiré interferometry, numerically with a finite element method (FEM), and analytically through a one-dimensional closed-form solution.^[4] They came to an interesting conclusion from their investigations, such that the laminated composite DLJs have considerable adherend shear deformations, which have been neglected in the theoretical modeling. They also observed that there are transverse normal stresses in the midplane of the adherend, and a strap moment in the midplane of the strap, as well as the longitudinal force. das Neves et al.^[5] developed an analytical model, taking into account the shear stress-free condition in the adhesive at the overlap ends. Kumar and Pandey conducted the two-dimensional (2D) and three-dimensional nonlinear (geometric and material) finite element (FE) analyses of adhesively bonded SLJs having modulus-graded bondline.^[6] In the study, the adhesives were modeled as an elasto-plastic multi-linear material, while the substrates were regarded as both linear elastic and bi-linear elasto-plastic material. They observed that the static strength is higher for joints with hybrid-adhesive bondlines compared to those with single adhesives in bondline. They also analytically showed that the in-plane global stiffness of the joint remains unaffected by the modulus gradation of the bondline adhesive. Hybrid-adhesive joints and their manufacturing were comprehensively reviewed by Safar et al.^[7] They reported that, although numerous studies have been conducted on the hybrid-adhesive joints, these studies are mainly limited to quasi-static loading conditions. They emphasized the necessity to investigate the behavior of hybrid-adhesive joints under dynamic loading conditions where the strain rate can considerably change the adhesive properties.

One of the first studies investigating the viscoelastic modeling of the adhesive joints was carried out by Delale and Erdogan.^[8] In the paper, the adhesively bonded joint problem was considered by assuming that the adhesive is a linear viscoelastic material. In their study, the stress distribution in the adhesive layer was calculated for three different external loads namely, membrane loading, bending, and transverse shear loading. Their results indicated that the peak value of the normal stress in the adhesive is not only consistently higher than the corresponding shear stress but also decays slower. Amidi and Wang established a viscoelastic analytical model for adhesively bonded SLJs

with functionally graded adherends.^[9] In the model, the adhesive layer was simulated as a three-parameter viscoelastic foundation using a standard linear solid (SLS) model to account for both creep and relaxation behaviors in the adhesive. Closed-form solutions of interface stresses were obtained and compared with FE analysis using ABAQUS. The effect of viscoelasticity of polymeric adhesives on the shear stress distribution in the adhesive layer of a DLJ under shear was studied by Shishesaz and Reza.^[10] The influence of the adhesive thickness, as well as the effect of adhesive viscosity and its shear modulus, on shear stress distribution in the adhesive overlap were also investigated. In the analytical modeling, the shear-lag model was used to obtain the governing differential equilibrium equations. These equations were solved using the residue theorem. The effect of viscoelasticity on the creep behavior of double-lap adhesively bonded joints was studied by Reza et al.^[11] In the article, Prony series was used to model the relaxation behavior of epoxy adhesive. According to their results, for an impulse load of 100 N, maximum shear stress in the adhesive layer was reduced to 38% of its initial value after almost 12 days and 79% of its initial value over a very long time.

To our knowledge, the viscoelastic behavior of the hybrid-adhesive joints was studied both analytically and numerically for the first time in the literature by Çay.^[12] In the dissertation, general differential equations of stress and strain were obtained in the Laplace domain, and transferred to the time domain by using the Fixed Talbot method. According to the results provided by Çay, the amount of the flexible adhesive used at the overlap ends remarkably determines the viscoelastic effects. Zehsaz et al.^[13] studied the creep behavior of the adhesively bonded SLJ using FEM. They showed that due to the creep straining, the stresses in joint corners decrease; however, creep causes shear and peel strains to increase in these areas, which this in turn may lead to separation. The authors suggest two methods to overcome and eliminate the problem of strain accumulation, which are increasing the layer thickness and using filleted joints, respectively. Özer and Çay developed a viscoelastic model for studying the viscoelastic behavior of a hybrid-adhesive joint, and investigated the effect of the hybrid-adhesive order on the distributions of the adhesive time-dependent shear stress.^[14] According to their results, shear stress reduction was observed at the ends of the overlap region with respect to time both for the hybrid-adhesive and mono-adhesive models, on the contrary, shear stress increased in the middle of the overlap region with respect to time for all type of models. Çay and Özer presented an extended version of their previous work,^[14] to develop viscoelastic semi-analytical and numerical models by considering the moment effects.^[15] They showed that optimizing the hybrid ratio for the hybrid-adhesive joints is the primary factor when considering the viscoelastic effects. They also concluded that the time-dependent nature of the adhesives significantly reduces and optimizes the stress and strain levels for the mono- and hybrid-adhesive

bondlines. Karlsson conducted an experimental study to determine the viscoelastic parameters of six different adhesives.^[16] The parameters were then used in combination with a Maxwell-Wiechert (MW) viscoelastic material model describing the viscoelasticity mathematically.

This study is dedicated to investigate the time-dependent behavior of the hybrid adhesively bonded DLJs. (The word hybrid is used here to represent the use of two different adhesives together in the overlap region.) For this purpose, stress and strain differential equations of a hybrid-adhesive DLJ were derived fully analytically in the Laplace domain. Then, a numerical inversion scheme was used to convert the shear stress and strain equations into the time domain. The analytical solution was also validated by using FEM. It is seen that the results of the proposed analytical model and the FEM are in a good agreement. In the last section, four different hybrid ratios were investigated, and compared with the mono-adhesive models. Effects of the different types of loads and adherend thicknesses were also studied. It was seen that using the hybrid-adhesive joints is a very effective way to reduce the shear stresses at the overlap ends. It was also observed that applying ramp load instead of the step load can be an effective method to reduce the peak shear stresses at the overlap ends. This research can lead to the proper selection of the hybrid ratio for the hybrid-adhesive DLJs by considering both the elastic and viscoelastic behaviors.

2. Joint geometry, materials, and modeling

Figure 1 shows the geometry and the dimensional parameters of the hybrid-adhesive DLJ. Adherends and adhesives are assumed to behave as linear elastic and linear viscoelastic, respectively. In Figure 1, subscripts “1, 2, 3”, and “a” correspond to the upper adherend, middle adherend, lower adherend, and

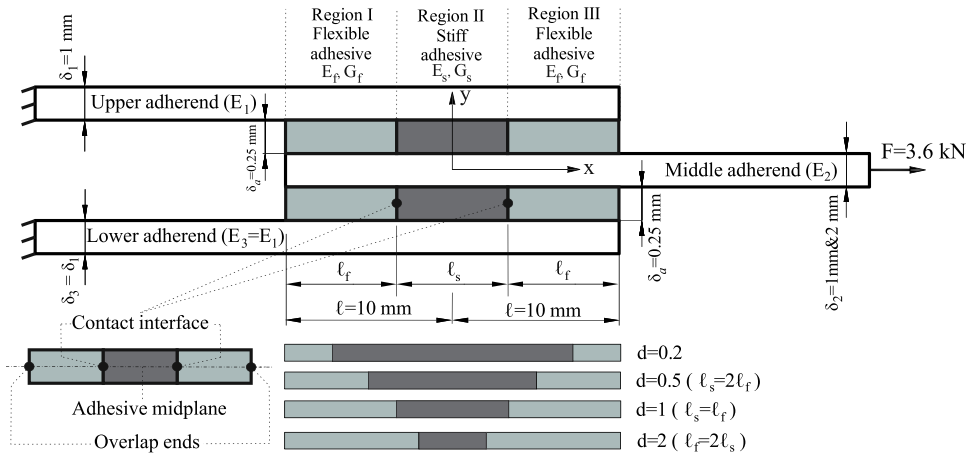


Figure 1. General view of the hybrid-adhesive DLJ.

adhesives, respectively. Adherend material is S235JR steel. Upper and lower adherends are completely identical by means of the material and geometric properties. Upper adherend, lower adherend, and adhesive layers have a constant thickness of $\delta_1=\delta_3=1$ mm and $\delta_a=0.25$ mm, respectively. Two types of middle adherend thicknesses have been investigated as $\delta_2=1$ mm and $\delta_2=2$ mm. The joint has a width of $b=40$ mm. Subscripts f and s correspond to the flexible and stiff adhesives, respectively. Therefore, the lengths of the flexible and stiff adhesives in the overlap are represented as ℓ_f and ℓ_s , respectively. The overlap length is also constant and 20 mm length, i.e. $2\ell = 2\ell_f + \ell_s$ (see Figure 1). Table 1 shows the elastic properties of the adherends and adhesives.^[14] In this study, SikaFast-5215 NT is selected as a flexible adhesive, and SikaPower-4720 is a stiff adhesive.

A parameter d named as the hybrid ratio is described as the ratio of the flexible adhesive to stiff adhesive lengths, defined as $d=\ell_f/\ell_s=0.2, 0.5, 1$, and 2 . Although the overlap is split into three regions, the flexible adhesive regions I and III are fully identical (see Figure 1).

The joint is subjected to a uniaxial tensile load from the right end of the middle adherend. The magnitude of the force is selected as 3.6 kN.^[14] The upper and the lower adherends are fixed from their left ends. DLJ tends to prevent bending moment due to its nature of symmetrical geometric properties shown in Figure 1. This paper attempts to investigate the shear stresses and strains that occurred in the adhesive layer.

The adhesives and viscoelastic parameters used in this study were taken from an experimental study in the literature.^[16] In the aforementioned study, it was reported that in the process of curve fitting and determining viscoelastic parameters, the experimental results could be most appropriately represented by the MW model. The viscoelastic behavior of the adhesives is therefore defined by a MW model with three units as shown in Figure 2.

A MW model with three units consists of a spring and three Maxwell chains in parallel. E_i and η_i ($i = 1, 2, 3$) correspond to the stiffness of the springs and viscous modulus of the adhesives, respectively. The long-term stiffness of the adhesive is represented by E_∞ . Table 2 summarizes the parameters of the spring and damper coefficients of the adhesives.^[14,16]

The thickness of the adhesives and adherends are very small considering the other dimensions of the geometry so the plane strain assumptions are taken

Table 1. Mechanical properties of the adherends and adhesives.^[14]

	Adherends S235JR (steel)	Flexible Adhesive SikaFast-5215 NT	Stiff Adhesive SikaPower-4720
Young's modulus (GPa)	207	0.25	1.9
Modulus of shear (GPa)	79.6	0.0912	0.695
Poisson's ratio	0.3	0.371	0.367
Tensile-Lap Shear strength (MPa)	-	10	14
Elongation at break (%)	26	200	3
Glass transition temperature ($^{\circ}\text{C}$)	-	60	73
Service temperature ($^{\circ}\text{C}$)	-	-40 ... +80	-30 ... +120

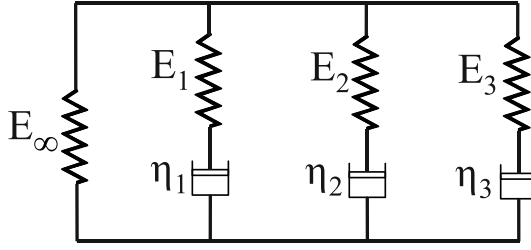


Figure 2. MW model with three units.

Table 2. Spring and damper coefficients for the adhesives.^[14,16]

	E_0 MPa	E_∞ MPa	E_1 MPa	E_2 MPa	E_3 MPa	η_1 MPa.s	η_2 MPa.s	η_3 MPa.s
SikaFast-5215 NT (Flexible adhesive)	467	81	151	140	94	104	1567	25190
SikaPower-4720 (Stiff adhesive)	2700	1917	249	301	233	183	3640	89794

into account while governing the analytical equations, and performing the FE analyses.

3. Analytical formulation of the problem

In this study, it is assumed that the vertical deformations of the adherends are negligible when compared with axial deformations. Hence, the shear stress along the adhesive thickness remains constant.^[17] On the other hand, a similar procedure of the shear-lag model is followed, and extended for governing the differential equations of the model.^[14] The shear-lag model only takes shear stresses into account for evaluation of the adhesively bonded lap joints. Although the shear-lag model is inadequate for evaluating the SLJs due to the ignoring of the bending moments, it is more applicable to the DLJs.^[17]

3.1. Mono-adhesive joint

The free-body diagram of an infinitesimal element is shown in [Figure 3](#). Upper and lower adherends are fully identical. The differential equations of the force and the displacement equilibriums of the upper and middle adherends can be obtained as:

$$\frac{\partial T_1(x, t)}{\partial x} + \tau_{xy}(x, t) = 0 \quad (1)$$

$$\frac{\partial T_2(x, t)}{\partial x} - 2\tau_{xy}(x, t) = 0 \quad (2)$$

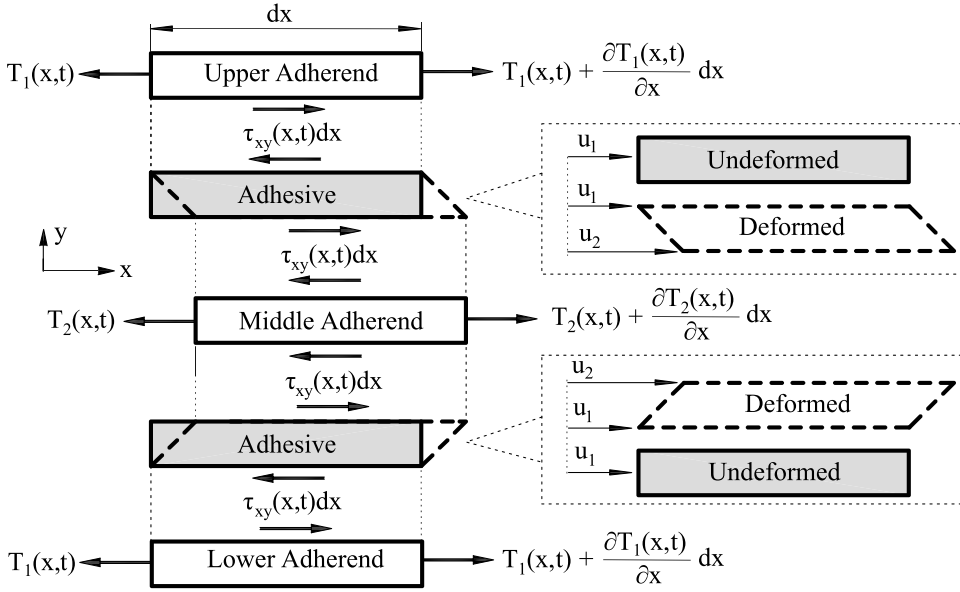


Figure 3. Free-body diagram of an infinitesimal element.

$$\varepsilon_1(x, t) = \frac{\partial U_1(x, t)}{\partial x} = \frac{T_1(x, t)}{E_1 \delta_1} \quad (3)$$

$$\varepsilon_2(x, t) = \frac{\partial U_2(x, t)}{\partial x} = \frac{T_2(x, t)}{E_2 \delta_2} \quad (4)$$

$$\gamma_{xy}(x, t) = \frac{U_2 - U_1}{\delta_a} \quad (5)$$

where the tensile force and axial strain for the upper adherend are T_1 and ε_1 , respectively. On the other hand, T_2 and ε_2 correspond to the force and axial strain of the middle adherend, respectively. u_1 and u_2 correspond to the horizontal displacements of the upper adherend and middle adherend, respectively.

The shear strain is represented by γ_{xy} . The viscoelastic definition of the shear stress for the adhesive layer can be expressed as follows^[18]:

$$\tau_{xy}(x, t) = \int_0^t G_a(t - \zeta) \frac{\partial \gamma_{xy}(x, \zeta)}{\partial \zeta} d\zeta \quad (6)$$

where G_a is the shear modulus of the adhesive, t is the time, and ζ is the variable of the integration. After applying the Laplace transform to Equation (6) one can obtain that:

$$\bar{\tau}_{xy}(x, s) = s\bar{G}(s)\bar{\gamma}_{xy}(x, s) \quad (7)$$

where s is the Laplace operator. The bar over any variable “—” indicates any parameter defined in the Laplace domain. The differential equations of the force and shear strain expressions can be rewritten in the Laplace domain as follows:

$$\frac{\partial \bar{T}_1(x, s)}{\partial x} + \bar{\tau}_{xy}(x, s) = 0 \quad (8)$$

$$\frac{\partial \bar{T}_2(x, s)}{\partial x} - 2\bar{\tau}_{xy}(x, s) = 0 \quad (9)$$

$$\bar{\gamma}_{xy}(x, s) = \frac{\bar{U}_2 - \bar{U}_1}{\delta_a} \quad (10)$$

Differentiating Equation (7) with respect to x yields:

$$\frac{\partial \bar{\tau}_{xy}(x, s)}{\partial x} = s\bar{G}(s) \frac{\partial \bar{\gamma}_{xy}(x, s)}{\partial x} \quad (11)$$

Then differentiating Equation (10) with respect to x , and substituting Equations (3) and (4) into it yields:

$$\frac{\partial \bar{\gamma}_{xy}(x, s)}{\partial x} = \frac{1}{\delta_a} \left[\frac{\bar{T}_2}{E_2 \delta_2} - \frac{\bar{T}_1}{E_1 \delta_1} \right] \quad (12)$$

Substituting Equation (12) into Equation (11) gives:

$$\frac{\partial \bar{\tau}_{xy}(x, s)}{\partial x} = \frac{s\bar{G}(s)}{\delta_a} \left[\frac{\bar{T}_2}{E_2 \delta_2} - \frac{\bar{T}_1}{E_1 \delta_1} \right] \quad (13)$$

Differentiating Equation (8) with respect to x , and substituting into Equation (13), we have:

$$\frac{\partial^2 \bar{T}_1}{\partial x^2} = -\frac{s\bar{G}(s)}{\delta_a} \left[\frac{\bar{T}_2}{E_2 \delta_2} - \frac{\bar{T}_1}{E_1 \delta_1} \right] \quad (14)$$

To eliminate one of the force term in Equation (14), one can write the total force equilibrium of the joint as follows:

$$\bar{T}_2 = \bar{F}(s) - 2\bar{T}_1 \quad (15)$$

Substituting Equation (15) into Equation (14), we can obtain the differential equation of the force equilibrium for the upper adherend in the following:

$$\frac{\partial^2 \bar{T}_1}{\partial x^2} - \frac{s\bar{G}(s)}{\delta_a} \left[\frac{2}{E_2 \delta_2} + \frac{1}{E_1 \delta_1} \right] \bar{T}_1 = -\frac{s\bar{G}(s)}{E_2 \delta_2 \delta_a} \bar{F}(s) \quad (16)$$

Equation (16), a second-order ordinary differential equation, can be rearranged in a general form as:

$$\frac{\partial^2 \bar{T}_1}{\partial x^2} - \alpha^2 \bar{T}_1 = \beta \quad (17)$$

where,

$$\begin{bmatrix} \alpha^2 \\ \beta \end{bmatrix} = \begin{bmatrix} \frac{s\bar{G}(s)}{\delta_a} \left[\frac{2}{E_2\delta_2} + \frac{1}{E_1\delta_1} \right] \\ -\frac{s\bar{G}(s)}{E_2\delta_2\delta_a} \bar{F}(s) \end{bmatrix} \quad (18)$$

The general solution of Equation (17) can be written as follows:

$$\bar{T}_1 = A \sinh(\alpha x) + B \cosh(\alpha x) - \frac{\beta}{\alpha^2} \quad (19)$$

Then, considering Equation (8), the shear stress expression can be obtained by differentiating Equation (19) with respect to x as follows:

$$\bar{\tau}_{xy} = -A\alpha \cosh(\alpha x) - B\alpha \sinh(\alpha x) \quad (20)$$

where A and B are the constants that are going to be evaluated using the boundary conditions (BCs).

3.2. Hybrid-adhesive joint

It can be seen from Figure 1 that a hybrid-adhesive joint consists of three individual regions named regions I, II, and III. Equations (19) and (20) will be adapted to evaluate the adherend forces and shear stresses region by region as follows:

$$\bar{T}_1^I = A_1 \sinh(\alpha^I x) + B_1 \cosh(\alpha^I x) - \frac{\beta^I}{(\alpha^I)^2} = > \{-\ell \leq x \leq -\ell_s/2\} \quad (21a)$$

$$\bar{T}_1^{II} = A_2 \sinh(\alpha^{II} x) + B_2 \cosh(\alpha^{II} x) - \frac{\beta^{II}}{(\alpha^{II})^2} = > \{-\ell_s/2 \leq x \leq \ell_s/2\} \quad (21b)$$

$$\bar{T}_1^{III} = A_3 \sinh(\alpha^I x) + B_3 \cosh(\alpha^I x) - \frac{\beta^I}{(\alpha^I)^2} = > \{\ell_s/2 \leq x \leq \ell\} \quad (21c)$$

$$\bar{\tau}_{xy}^I = -A_1 \alpha^I \cosh(\alpha^I x) - B_1 \alpha^I \sinh(\alpha^I x) = > \{-\ell \leq x \leq -\ell_s/2\} \quad (22a)$$

$$\bar{\tau}_{xy}^{II} = -A_2 \alpha^{II} \cosh(\alpha^{II} x) - B_2 \alpha^{II} \sinh(\alpha^{II} x) = > \{-\ell_s/2 \leq x \leq \ell_s/2\} \quad (22b)$$

$$\bar{\tau}_{xy}^{III} = -A_3\alpha^I \cos h(\alpha^I x) - B_3\alpha^I \sinh(\alpha^I x) = > \{\ell_s/2 \leq x \leq \ell\} \quad (22c)$$

Despite there are different regions as I and III at the overlap ends, the flexible adhesive regions are fully identical. Hence, the representative equations that belong to these regions have the same parameters of α and β .

We need six BCs to calculate the coefficients A_i and B_i ($i = 1, 2, 3$) in Equations (21) and (22). **Figure 4** shows the BCs of the hybrid-adhesive DLJ.

Considering **Figure 4**, the related BCs for the problem can be written as below:

$$\bar{T}_1^I \Big|_{x=-\ell} = \frac{\bar{F}(s)}{2} \quad (23a)$$

$$\bar{T}_1^{III} \Big|_{x=\ell} = 0 \quad (23b)$$

$$\bar{T}_1^I \Big|_{x=-\ell_s/2} = \bar{T}_1^{II} \Big|_{x=-\ell_s/2} \quad (23c)$$

$$\bar{T}_1^{II} \Big|_{x=\ell_s/2} = \bar{T}_1^{III} \Big|_{x=\ell_s/2} \quad (23d)$$

$$\frac{\bar{\tau}_{xy}^I}{s\bar{G}^I(s)} \Big|_{x=-\ell_s/2} = \frac{\bar{\tau}_{xy}^{II}}{s\bar{G}^{II}(s)} \Big|_{x=-\ell_s/2} \quad (23e)$$

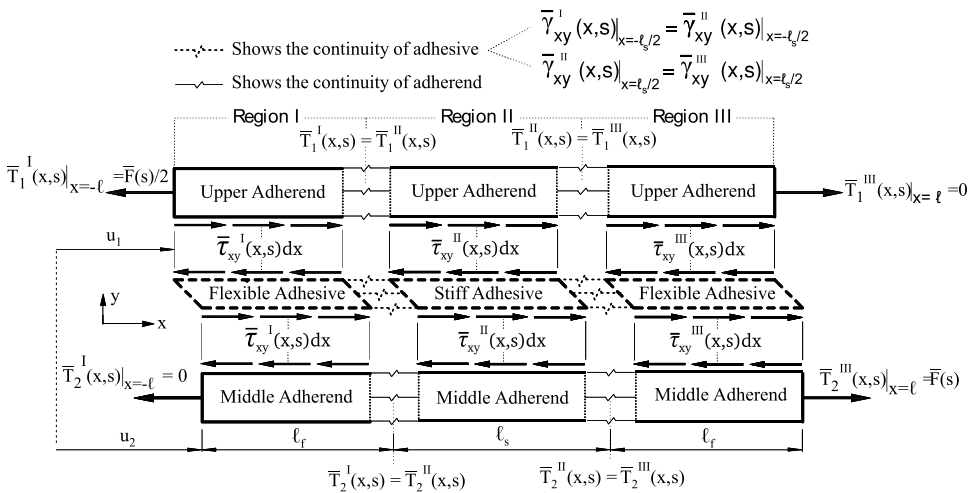


Figure 4. BCs of the DLJ given on the free-body diagram.

$$\left. \frac{\bar{\tau}_{xy}^{II}}{s\bar{G}^{II}(s)} \right|_{x=\ell_s/2} = \left. \frac{\bar{\tau}_{xy}^{III}}{s\bar{G}^{I}(s)} \right|_{x=\ell_s/2} \quad (23f)$$

The load F is subjected from the right end of the middle adherend, and has two types (see [Figure 5a](#)). One of the loading types is step load, and the other one is ramp load. In the Laplace domain, the step load can be expressed as follows:

$$\bar{F}(s) = \frac{F}{s} \quad (24)$$

Similarly, a ramp load can be defined as:

$$\bar{F}(s) = \frac{F}{t_p} \left(\frac{1 - e^{-t_p s}}{s^2} \right) \quad (25)$$

where t_p corresponds to the retardation time which the load F reaches from zero to its maximum value. Shear modulus of the adhesives can be defined in the Laplace domain as follows^[14]:

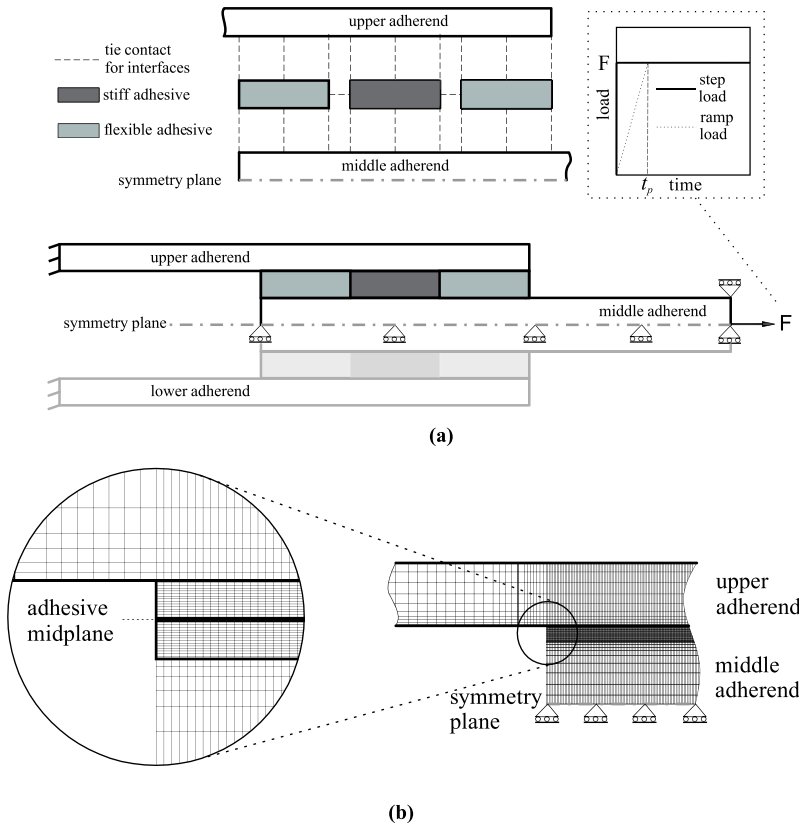


Figure 5. Details of FEM of the DLJ: (a) BCs, tie contacts between surfaces, and load types; (b) mesh model.

$$\overline{G}(s) = \left[G_{\infty} + \sum_{j=1}^3 \frac{sG_j}{s + \frac{1}{\eta_j/E_j}} \right] \frac{1}{s} \quad (26)$$

Inverse Laplace conversion of a complicated function is often a difficult stage of the solution of a time-dependent problem. Therefore, a numerical inversion method is used to convert the Equation (22) obtained analytically into the time domain.^[19] In the authors' previous studies, the Fixed-Talbot method was used to overcome the inversion problem.^[14,15] In this study, a considerable amount of CPU time has been saved without compromising the precision of the solution by using the principles described in the reference.^[19] An inverse Laplace transform of a function can be defined as^[19]:

$$f(t) = L^{-1}\{F(s)\} = \frac{1}{2\pi j} \int_{\theta-j\infty}^{\theta+j\infty} F(s)e^{st} ds \quad (27)$$

where $s=\theta+j\infty$ is defined as the integration variable. Exponential expression in Equation (27) can be simplified as in the following expression:

$$e^{st} \doteq E_c(st, \mu) = \frac{e^{\mu}}{2 \cosh(\mu - st)} = \frac{e^{st}}{1 + e^{-2\mu} e^{2st}} \quad (28)$$

A convergent MacLaurin series can then be obtained when $\mu > \theta t$ as follows:

$$E_c(st, \mu) = e^{st} + \sum_{n=1}^{\infty} (-1)^n e^{-2n\mu} e^{(2n+1)st} \quad (29)$$

Expression in the sum operator corresponds to the error of the approximation. The error can be reduced by using the suitable selection of parameter μ . Then, using Equations (27), (28), and (29), approximation can be defined as follows:

$$\begin{aligned} f(t) \doteq f_c(t, \mu) &= \frac{1}{2\pi j} \int_{\theta-j\infty}^{\theta+j\infty} F(s) E_c(st, \mu) ds = f(t) + \sum_{n=1}^{\infty} (-1)^n e^{-2n\mu} [f(2n+1)t] \\ &= f(t) - e^{-2\mu} f(3t) + e^{-4\mu} f(5t) - \dots = f(t) + \epsilon_c(t, \mu) \end{aligned} \quad (30)$$

where the $\epsilon_c(t, \mu)$ corresponds to the error $E_c(st, \mu)$ can be expressed as an infinite sum of rational functions in st . Hence, the inversion problem can be rewritten in the following form by making some manipulations:

$$f_c(t, \mu) = \frac{e^{\mu}}{2j} \sum_{n=0}^{\infty} (-1)^n (n + 1/2) I_n \quad (31a)$$

where,

$$I_n = \int_{\theta-j\infty}^{\theta+j\infty} \frac{F(s)}{(n+1/2)^2\pi^2 + (\mu-st)^2} ds = \int_{\theta-j\infty}^{\theta+j\infty} \frac{F(s)}{G_n(s)} ds = \int_{\theta-j\infty}^{\theta+j\infty} H_n(s) ds \quad (31b)$$

Due to the assumption of $\lim_{|s| \rightarrow \infty} F(s) = 0$, I_n will equal to the sum of the residues of $H_n(s)$ multiplied by $2\pi j$ as follows:

$$I_n = -2\pi j [res|_{s=s_1} H_n(s) + res|_{s=s_2} H_n(s)] = -2\pi j \left[\frac{F(s_1)}{G'_n(s_1)} + \frac{F(s_2)}{G'_n(s_2)} \right] \quad (32)$$

At last, the inverse Laplace transformation can be completed as follows:

$$f_c(t, \mu) = -\frac{e^\mu}{t} \sum_{n=0}^{\infty} (-1)^n Im \left\{ F \left[\frac{\mu}{t} + j(n+1/2) \frac{\pi}{t} \right] \right\} \quad (33)$$

4. Finite element model

The analytical model was tested by using a 2D FEM via a commercial code ABAQUS. A user material code (UMAT) was run at the background of the FE analyses for modeling the linear viscoelastic behavior of the adhesives. The contact surfaces were defined by using the tie contact option. Making the tie contact between surfaces means that the surfaces are connected completely and rigidly. The contacts, the boundary conditions, the load applied to the middle adherend, load types, and mesh details are shown in [Figure 5](#). A symmetrical FE model was developed to reduce the cost of the CPU time. A virtual plane of symmetry was then defined to reflect the symmetrical BCs, and shown in [Figure 5a](#).

It is assumed that perfect bonding is achieved between the contact interfaces (see [Figure 1](#)) of the adhesives and between the adhesive-adherend surfaces. Step load was applied to compare the analytical model with the numerical one. Firstly, comparisons of the analytical and FE models of a DLJ were done for the mono- and hybrid-adhesive model with $d=2$.

All of the adherends and adhesives were meshed using the four-node plane strain element called CPE4, and the time increment was set as fixed.^[14] (Analyses were also made considering the state of plane stress and it was seen that the results were very similar to each other.) The number of nodes and elements was increased in the adhesive layer due to the performing of some convergence tests. After optimization of the element size, the meshing structure of the adhesive region was shaped as a 30×800 rectangular matrix, and one of the element size in the adhesive region is $0.00833 \text{ mm} \times 0.025 \text{ mm}$. Meshings near the edges of the adhesive-adherend interfaces were refined to guarantee the precision of the results. FEM has a totally of 41,200 elements and

43,835 nodes for both the mono- and hybrid-adhesive models. Shear stress distributions were obtained along a virtual line through the adhesive-midplane as shown in Figure 5b. Hybrid-adhesive joints with different hybrid ratios, different middle adherend thicknesses, and effects of the load types were investigated analytically in the subsequent section.

5. Results and discussion

5.1. Viscoelastic analysis of the mono-adhesive joint: step loading case

Before examining the hybrid-adhesive joints, a preliminary examination for the mono-adhesive joints is investigated. The term “*mono-flexible*” to be used in the figures refers to the use of only ductile adhesive throughout the bonding overlap. A similar situation applies to “*mono-stiff*” one.

Figure 6 shows the analytical and numerical comparison of the maximum shear strain for the mono-flexible and mono-stiff DLJs. It is seen from Figure 6 that the relaxation of the adhesives was fully stable about the time of $t=10000$ s. Numerical results predict higher peak shear-strain values for the mono-flexible joints; on the contrary, the analytical results predict higher peak shear-strain values for the mono-stiff one. Figure 6a shows that the increase in the shear strain from $t=0$ to the relaxation time of $t = 10000$ s is approximately 391.5% by considering the analytical results. The numerical result is approximately 5.7% higher than the analytical result at the time of relaxation. On the other hand, Figure 6b depicts that the increase in the shear strain for the mono-stiff joint is approximately 22.7% by considering the analytical results. The analytical result is approximately 7.7% higher than the numerical result at the time of relaxation.

Figure 7 depicts the analytical and numerical results of the shear stress distribution along the overlap for $t = 0$ and $t = 10000$ s. A good match between

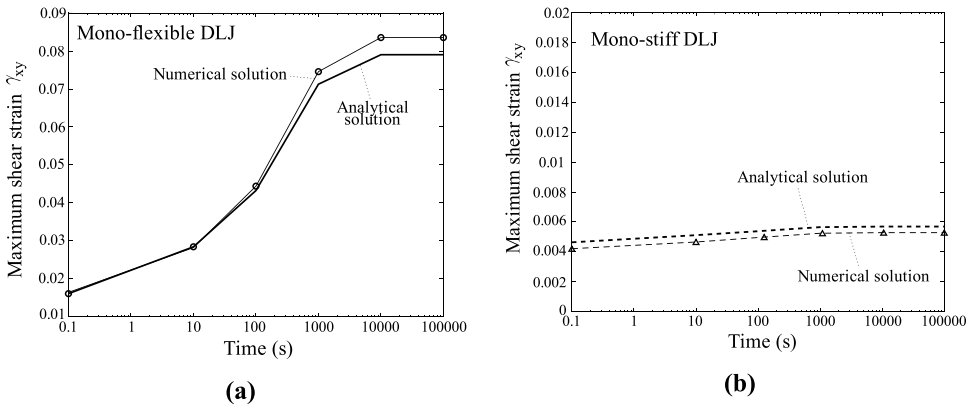


Figure 6. Analytical and numerical comparison of the peak shear strains for step loading case: (a) the mono-flexible DLJ; (b) the mono-stiff DLJ.

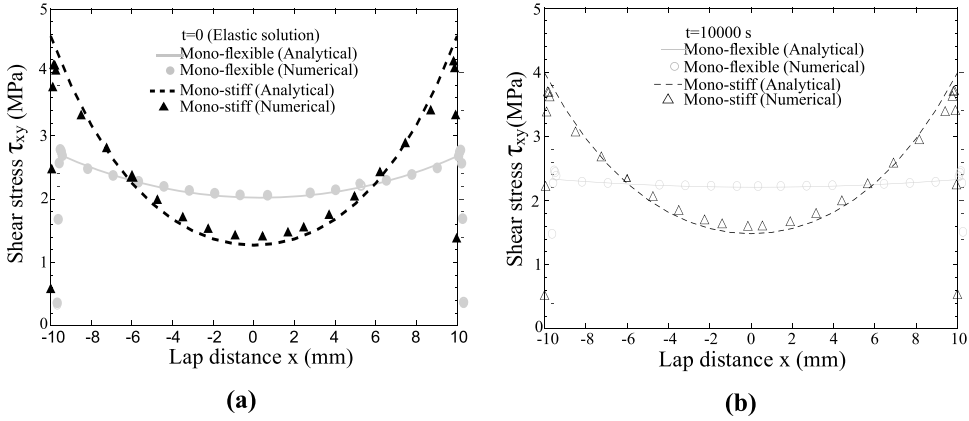


Figure 7. Analytical and numerical comparison of the shear stresses occurred in the mono-flexible and mono-stiff joints for step loading case: (a) $t=0$; (b) $t=10000$ s.

the analytical and numerical results was observed. The maximum shear stresses occur at the overlap ends as expected. It is seen from Figure 7 that the numerical model tends to indicate the zero shear stress condition at the overlap ends.

5.2. Preliminary viscoelastic analysis of the hybrid-adhesive joint with the hybrid ratio of $d = 2$: step loading case

Examining the mono-adhesive joints by using the analytical and numerical methods was an introduction before moving to the hybrid-adhesive joints. In this section, a hybrid-adhesive joint with a hybrid ratio of $d = 2$ was investigated using the analytical and numerical models. Hybrid-adhesive joint with hybrid ratio $d = 2$ is a special joint because the stiff adhesive length ℓ_s is half the length of the flexible adhesive ($\ell_f / 2$). Therefore, the aim of choosing this hybrid ratio is to demonstrate how a small amount of the stiff adhesive in the hybrid overlap affects the viscoelastic behavior.

Figure 8 shows a comparison of the maximum shear strains of the hybrid-adhesive joint with $d=2$ in terms of the analytical and numerical results. As seen in Figure 8, numerical results overestimate the shear strain for the hybrid-adhesive joint. Before $t=1000$ s, maximum shear strain increases remarkably. Increase rate of the maximum shear strain is getting slower between the times of $t=1000$ s and $t=10000$ s. It becomes fully stable after $t=10000$ s. Consequently, the relaxation time can be assumed as $t=10000$ s for the hybrid-adhesive joint. One can obtain from Figure 8 that the increase in the shear strain is approximately 80.5% by considering the analytical results. Although the increase in the shear strain is approximately four times higher than that of the mono-stiff joint, it is approximately five times lower than that

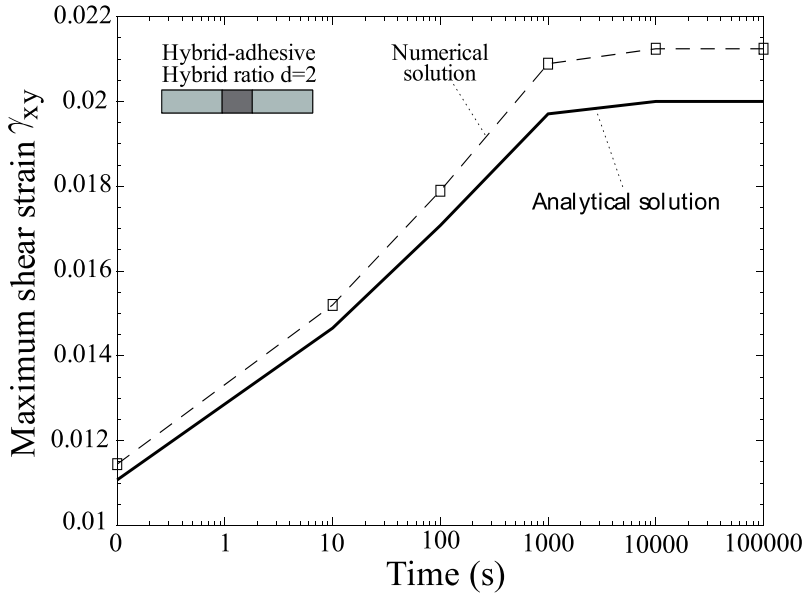


Figure 8. Analytical and numerical comparison of the peak shear strains of the hybrid-adhesive joint with $d=2$ for step loading case.

of the mono-flexible one. Hence, the effects of a small amount of the stiff adhesive inside the hybrid overlap on viscoelastic behavior can be seen clearly.

The shear stress distribution along the overlap region is plotted in Figure 9 by considering both the analytical and numerical results. It can be seen from Figure 9 that a perfect match is obtained between the results. Figure 9 shows that the shear stress of the stiff adhesive region increases; conversely, the shear stress reduces at the ends of the overlap as time progresses from $t = 0$ to $t = 10000$ s. Figure 9 also shows that the analytical solution predicts the shear stresses as 1.87 and 0.59 MPa at the overlap ends for the times of $t = 0$ and $t = 10000$ s, respectively. However, the shear stresses occurred as 5.94 and 9.60 MPa in the contact interfaces for the times of $t = 0$ and $t = 10000$ s, respectively. Despite the shear stress reducing 68.5% at the overlap ends, it increases by 61.6% at the contact interfaces.

5.3. Viscoelastic investigation of the hybrid-adhesive DLJs: step loading case

In this section, four different hybrid-adhesive joints with $d=0.2$, 0.5, 1, and 2 will be investigated in detail. Hybrid-adhesive joints are also compared with the mono-adhesive models. Considering the compatibility of the analytical results with the numerical ones in the previous sections, only the analytical results will be used in the rest of the study (A study similar to this study was also conducted for the SLJ.^[14] The parameters of the two models are exactly the same, only the DLJ model was applied with a load twice the load used in the SLJ. It was seen that

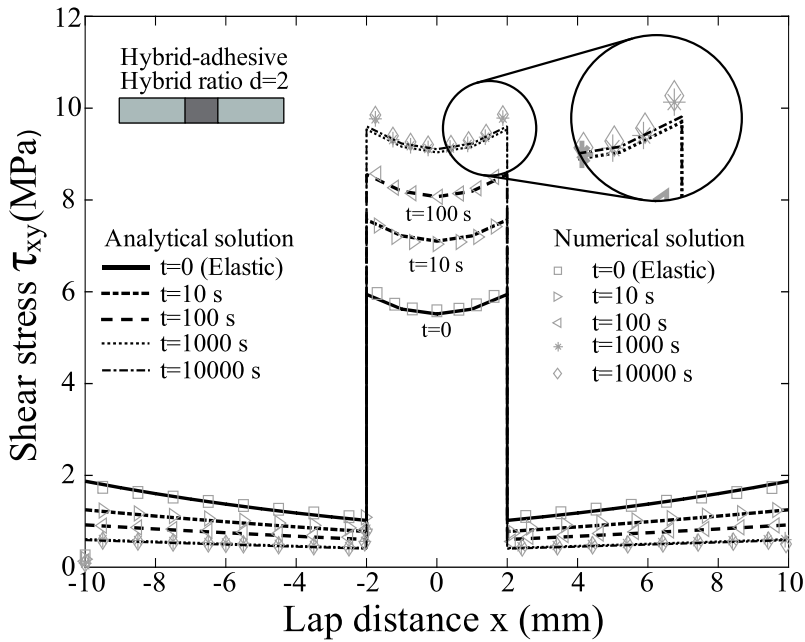


Figure 9. Analytical and numerical comparison of the shear stresses of the hybrid-adhesive joint with $d=2$ at the times of $t=0, 10, 100, 1000$, and $10,000$ s for step loading case.

the results obtained from the two models were found to be compatible.). The hybrid ratios $d=0.5$ and $d=2$ have been selected especially when compared to the other ratios. The reason for choosing these two ratios is that in one of the ratios, the adhesive length at the overlap ends is maximum. The flexible adhesive length is two times longer than the stiff adhesive length for $d=2$, i.e. $8-4-8$ mm, the stiff adhesive length is two times longer than that of the flexible adhesive for $d=0.5$, i.e. $5-10-5$ mm.

Figure 10 shows the peak shear strains for all of the mono- and hybrid-adhesive joints. Mono-flexible and mono-stiff models are also plotted in Figure 10 to see the effect of the hybrid-adhesive order more clearly. The peak shear strains of the hybrid-adhesive models are considerably lower than that of the mono-flexible one. A 28.2% increase in the peak shear strain was observed for $d=0.2$ between the times of $t=0$ and $t=10000$ s. However, a 38.9% increase in the peak shear strain was observed for $d=0.5$ within the same time intervals.

After examining the peak shear strains for the hybrid-adhesive joints, shear stress investigations were also performed, and summarized in Figure 11 both for the times of $t=0$ and $t=10000$ s.

As the hybrid ratio increases, the increase of the shear stress gets higher at the contact interfaces. For instance, it is evaluated from Figure 11a that the shear stresses at the contact interfaces of the hybrid-

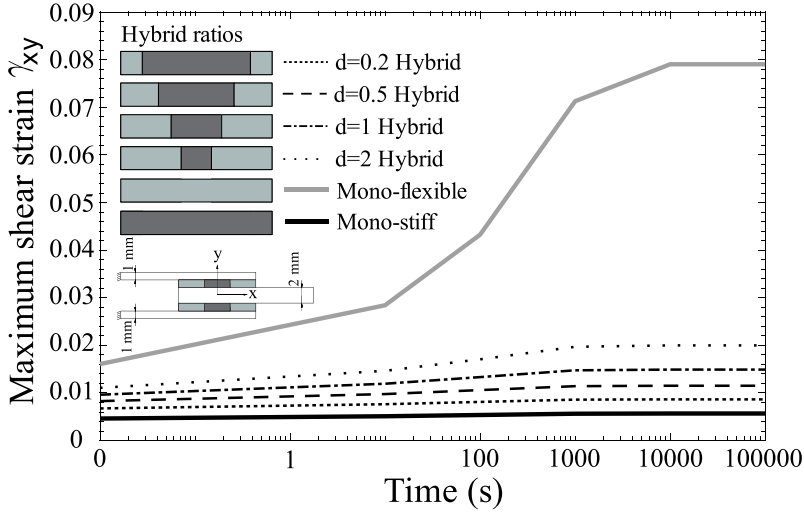


Figure 10. Comparison of the peak shear strains for the mono- and hybrid-adhesive models with respect to time for step loading case.

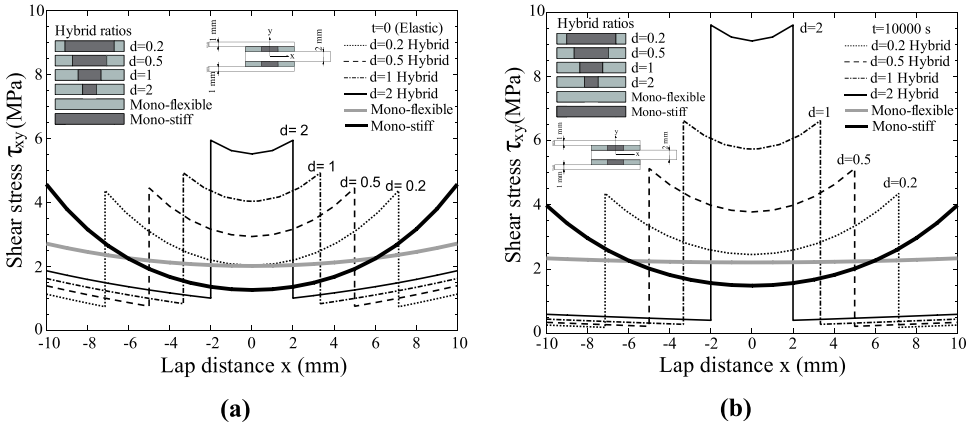


Figure 11. Comparison of the shear stresses for the hybrid- and mono-adhesive DLJs for the middle adherend thickness of $\delta_2=2$ mm for step loading case: (a) $t=0$; (b) $t=10000$ s.

adhesive joint with $d=2$ are approximately 5.94 and 9.60 MPa for $t=0$ and $t=10000$ s, respectively. However, shear stresses at the contact interfaces for $d=0.5$ are approximately 4.46 and 5.14 MPa for $t=0$ and $t=10000$ s, respectively. While the increase in the shear stress is 61.6% for $d=2$, the increase for the hybrid-adhesive joint with $d=0.5$ is 15.2%. Minimum shear stresses at the adhesive interfaces are also seen in the hybrid-adhesive joint with $d=0.2$. Shear stresses at the overlap ends are lower than those of the mono-adhesive models for all types of the hybrid-adhesive joints. Hybrid-adhesive joint with $d=2$ has the maximum shear stresses occurring at the contact interfaces by comparing

with the other hybrid ratios. As seen from Table 1, flexible adhesive in the hybrid-adhesive joint with $d=2$ approaches its shear strength limit.

5.4. Viscoelastic investigation of the hybrid-adhesive DLJs: effect of the middle adherend thickness for step loading case

In the examinations conducted above, the middle-adherend thickness was twice the upper and lower adherend thicknesses, and therefore 2 mm. In order to examine the effect of the change in the middle-adherend thickness on stresses, the middle adherend thickness was taken as 1 mm and the analyses were repeated. When the thickness of the middle adherend is chosen as $\delta_2=1$ mm, an unbalanced stress distribution is obtained both for the contact interfaces and overlap ends. Figure 12 depicts the shear stress distribution both for the hybrid- and mono-adhesive joints for $t=0$ and $t=10000$ s for the middle adherend thickness of $\delta_2=1$ mm. As seen in Figure 12, the maximum shear stresses occur at the right contact interface for the hybrid-adhesive joints.

A remarkable result for $\delta_2=1$ mm hybrid-adhesive with $d=2$ is observed for $t=10000$ s. In spite of the peak shear stress obtained from Figure 11b is 9.60 MPa for hybrid-adhesive joint with $d=2$ and $\delta_2=2$ mm, the peak shear stress obtained from Figure 12b is 10.12 MPa for hybrid-adhesive joint with $d=2$ and $\delta_2=1$ mm.

It can be concluded from Figures 10, 11, and 12, that using the hybrid adhesive joints is a very effective way to reduce the shear stresses at the overlap ends. Furthermore, the joint parameters like the adherend thicknesses, hybrid ratio, etc., must be taken into account for the proper selection of the hybrid-adhesive joints.

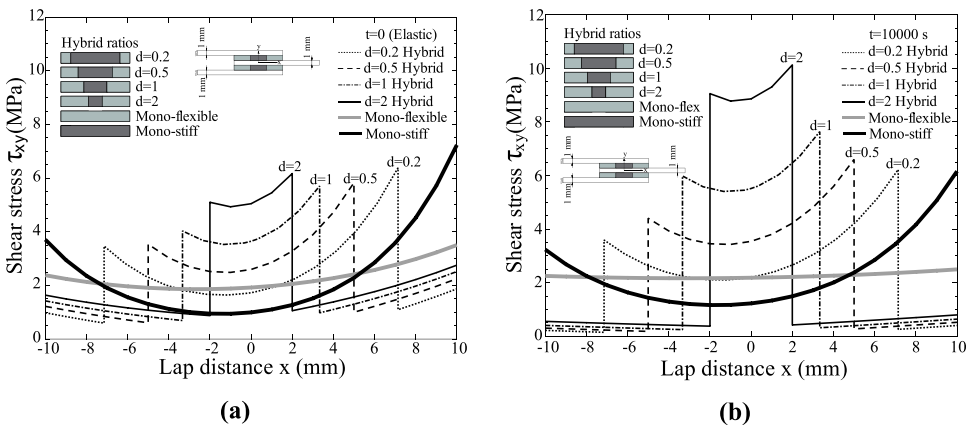


Figure 12. Comparison of the shear stresses for the hybrid- and mono-adhesive DLJs for the middle adherend thickness of $\delta_2=1$ mm for step loading case: (a) $t=0$; (b) $t=10000$ s.

5.5. Viscoelastic investigation of the hybrid-adhesive DLJs: ramp loading case

Not only the appropriate selection of the hybrid ratio for the hybrid-adhesive joints but also the loading effect may be an effective way to reduce the peak shear stresses in the overlap. Therefore, the effects of the loading type are also investigated. A ramp load already defined in Equation (25) is applied to the mono- and hybrid-adhesive joints. Mono-adhesive models were investigated first. Hence, Figure 13 shows the effect of the ramp load on the mono-adhesive joints for the retardation times of $t_p=1, 8, 16$, and 32 s. Step load results are also plotted in the figures. Applying a ramp load to the middle adherend means that the load reaches its maximum at the end of the retardation time. One can conclude from Figure 13 that the increase in the retardation time decreases the peak shear stresses. For example, applying ramp load to the mono-flexible joint with the $t_p=8$ s and $t_p=32$ s instead of the step load decreases the peak shear stresses by approximately 6.8% and 9.3%, respectively. Besides, a 5.2% and 7.5% decrease in the peak shear stresses are observed approximately for the mono-stiff joint for $t_p=8$ s and $t_p=32$ s, respectively.

Finally, in order to see the effect of the ramp load on the shear stresses for the hybrid-adhesive joints, Figure 14 is plotted. In this figure, the effects of the ramp load for the hybrid-adhesive joints with the minimum ($d=0.2$) and the maximum ($d=2$) hybrid ratios are shown. Figure 14b and 14d show the shear stresses at the contact interfaces. It can be concluded that the step load or ramp load condition do not change the peak shear stresses at the contact interfaces. On the contrary, peak shear stresses are reduced considerably at the overlap ends. It can be obtained from Figure 14a that applying ramp load with $t_p=8$ s and $t_p=32$ s instead of the step load for the hybrid-adhesive joint with $d=0.2$, reduces the peak shear stresses 32.6% and 46.2%, respectively. On the other hand, it can be obtained from Figure 14c that applying ramp load with $t_p=8$ s and $t_p=32$ s instead of the step load for the hybrid-adhesive joint with $d=2$, reduces the peak shear stresses 24.3% and 35.8%, respectively. Considering the

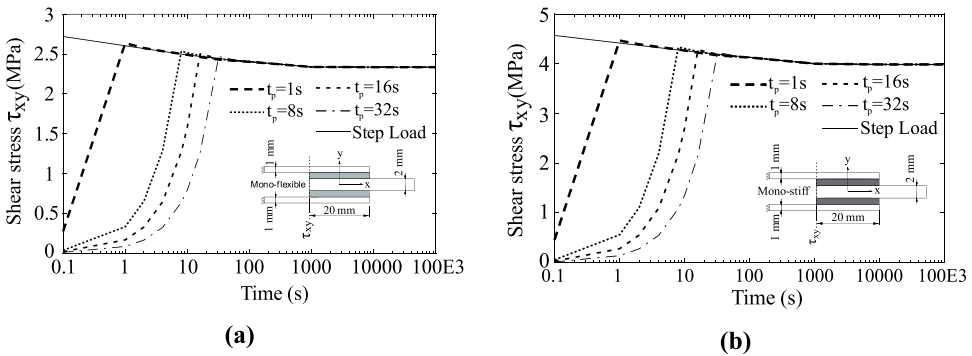


Figure 13. Effects of the ramp load on the maximum shear stresses for the mono-adhesive DLJs: (a) the mono-flexible model; (b) the mono-stiff model.

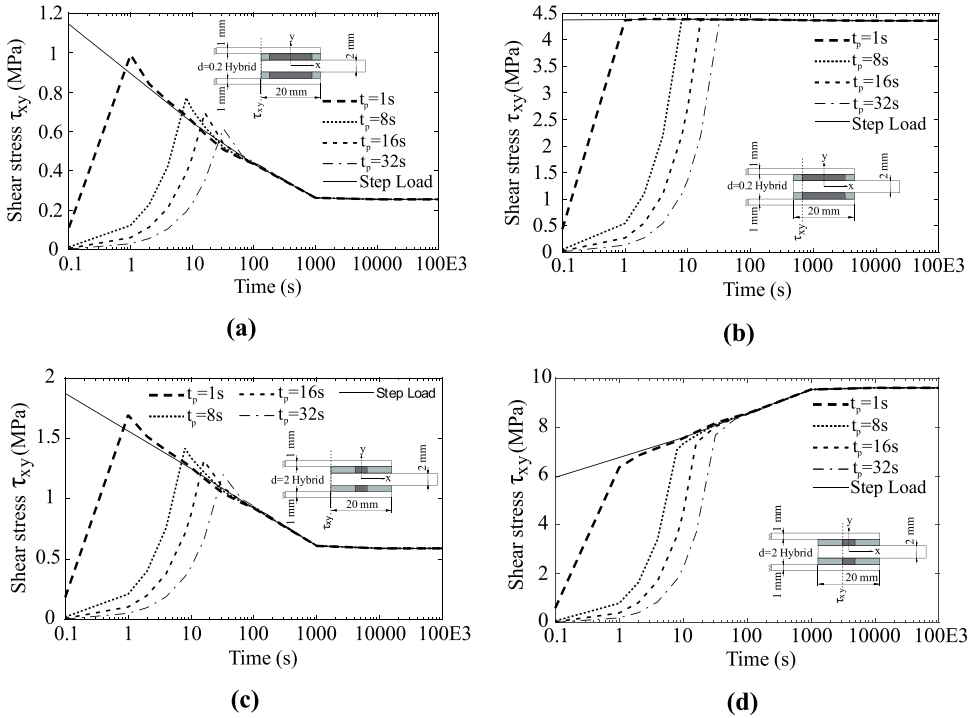


Figure 14. Effects of the ramp load on the shear stresses for the hybrid-adhesive DLJs: (a) $d=0.2$ at the overlap ends; (b) $d=0.2$ at the contact interfaces; (c) $d=2$ at the overlap ends; (d) $d=2$ at the contact interfaces.

mono- and hybrid-adhesive joints, it is observed that the hybrid-adhesive joints are more sensitive to the applying ramp load instead of step load at the overlap ends.

6. Conclusions

In this study, the shear stresses and strains developed in both mono- and hybrid-adhesive DLJs were investigated by taking into account the viscoelastic effects. Equilibrium and differential equations were derived analytically in the Laplace domain. Then, a numerical method was used to transfer the relevant expressions to the time domain. Analytical and numerical results were compared with each other, and the results were in a good agreement.

The obtained results showed that using even a small amount of rigid adhesive along the hybrid-adhesive overlap considerably restricts the effects of viscoelasticity. It was seen that, if all the adherends have the same thickness and material type, there is an increase in the peak shear stresses in both mono- and hybrid-adhesive DLJs by considering the case of the middle adherend thickness is twice the upper/lower adherend thickness. It is concluded that using the hybrid-adhesive joints is a reliable technique to

reduce the shear stresses at the overlap ends. Furthermore, applying ramp load instead of step one is an auxiliary way to reduce the peak shear stresses at the overlap ends. It is seen that the changes in the peak shear stresses occurring at the overlap ends of the hybrid-adhesive joints depending on the load type are more pronounced than ones in the mono-adhesive overlaps.

Considering the results obtained within the scope of this study, the hybrid ratio optimization should be performed not only according to the elastic analysis results, but instead by taking into account the effects of parameters such as viscoelastic effects, thicknesses of the adherends, and load type.

The analytical model developed within the scope of this study can be easily adapted to the case where adherends are anisotropic materials. Moreover, it seems that the relevant analytical model is open to modification to perform nonlinear viscoelastic modeling of adhesives.

Disclosure statement

No potential conflict of interest was reported by the authors.

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